



PlanetScope Basic Analytic: Quality Processes Review

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1 Introduction

1.1 Scope

This document summarises a Data Product Quality Processes Review for the PlanetScope Basic Analytic data product as part of the EODH Data Product Quality Process [AD-1]. The purpose is to demonstrate that the product has been developed and generated following processes that adhere to best practices.

To achieve this, the EODH Data Product QA Process adheres to the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects of Data Product quality – awarding grades of *Basic*, *Good*, *Excellent*, or *Ideal*. Such assessments are reported with a maturity matrix – this is presented in Section 2. The rest of the document provides the underpinning evidence justifying the awarded grades. This assessment has been completed by the EODH QA Service on behalf of the Data Product provider.

The result of this evaluation, referred to as a *QA Check Result*, will be added as an annotation to the EODH Catalogue for user interaction.

1.2 Changes Log

Issue	Date	Description
0.1	25/03/2025	Initial draft for inclusion in pathfinder demonstration.
1.01.0	22/02/2026	Updated draft for pre-publication review. Updates include: • Updating product name to match PlanetScope naming convention to ensure clarity • Updated Ancillary Data grade from “Basic” to “Good” • Updated Calibration Algorithm grade from “Good” to “Excellent” • Minor formatting updates

1.3 Applicable Documents

[AD-1] EODH QA Service, ‘Data Product Quality Assurance Process’, 2025.

1.4 Reference Documents

[RD-1] Hunt et al., ‘Earth Observation Mission Quality Assessment Framework’ v2.2, EDAP Project Document, 2021. [Available Online](#).

[RD-2] Planet L1 Data Quality QA 2021 Report, 2021

[RD-3] Understanding PlanetScope Instruments: [Understanding PlanetScope Instruments](#)

[RD-4] PlanetScope Constellation : [PlanetScope Constellation - Geospatial Intelligence](#)

[RD-5] PlanetScope Product Specifications, [PSScene Imagery Product Spec FINAL | June 2021](#), June 2021

- [RD-6] PlanetScope Product Specifications, [PSScene Imagery Product Spec FINAL | December 2023](#), December 2023
- [RD-7] Application Schema for EO Products OGC GML 3.1.1 Version 0.9.1
- [RD-8] PlanetScope Overview : [PlanetScope Overview](#)
- [RD-9] On-Orbit Radiometric Calibration of the Planet Satellite Fleet, [Radiometric Calibration White paper](#), January 2022
- [RD-10] JCGM, "Evaluation of measurement data - Guide to the expression of uncertainty in measurement," *JCGM*, vol. 100, 2008, Accessed: Feb. 06, 2018. [Online]. Available: https://www.bipm.org/utils/common/documents/jcgm/JCGM_100_2008_E.pdf.
- [RD-11] Planet Imagery Product Specifications : [combined-imagery-product-spec-final-august-2019.pdf](#), August 2019

1.5 Acronyms & Abbreviations

EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance

2 Review Summary – Product Maturity Matrix

The *Data Product Quality Processes Review* is aimed at demonstrating that a given Data Product is developed and generated following processes that adhere to best practices.

To achieve this, the EODH *Data Product QA Process* adheres to the ESA/NASA EO Product QA Framework [RD-1]. Such assessments are reported in a maturity matrix style, where each box represents an area of assessment for which underpinning requirements are defined – the box colour represents the achieved grade, *Basic*, *Good*, *Excellent*, or *Ideal*. Additionally:

- *Not Assessed* means this aspect of data quality was not assessed at this time.
- *Not Assessable* means the assessment of this aspect of data quality was not possible due to a lack of available evidence.
- The padlock symbol means that the supporting evidence reviewed is not publicly available.

The *Data Product Quality Processes Review* for PlanetScope Basic Analytic is presented in Figure 1. The rest of the document provides the underpinning evidence justifying the awarded grades, as follows:

- Section 3 provides the evidence for the Product Information column.
- Section 4 provides the evidence for the Metrology column.
- Section 5 provides the evidence for the Product Generation column.

Data Product Quality Processes Review			Key
Product Information	Metrology	Product Generation	
Product Details	Sensor Calibration & Characterisation	Calibration Algorithm	Not Assessed
Availability & Accessibility	Geometric Calibration & Characterisation	Geometric Processing	Not Assessable
Product Format, Flags & Metadata	Metrological Traceability Documentation		Basic
User Documentation	Uncertainty Characterisation		Good
	Ancillary Data		Excellent
			Ideal
			🔒 Not Public

Figure 1 - Data Product Quality Processes Review Maturity Matrix for PlanetScope Basic Analytic

3 Product Information

Product Details	
Grade: Excellent	
Justification	<i>Reasonable level of information available publicly, although across various references. [RD-2, RD-3, RD-4, RD-5, RD-6]</i>
Product Name	<i>PlanetScope Ortho Analytic 8B Scene</i>
Sensor Name	<i>PSB.SD – SuperDove</i>
Sensor Type	<i>VNIR multi-channel spectrometer (8 bands)</i>
Mission Type	<i>Constellation – ~100 satellites</i>
Mission Orbit	<i>Sun Synchronous</i>
Product Version Number	<i>Included in product metadata – e.g. 4.1.4</i>
Product ID	<i>YYYYMMDD_HHMMSS_cS_satid_3B_AnalyticMS_8b_clip.tif</i> - Where YYYYMMDD is the date of measurement, - HHMMSS is the time of measurement, - cS is the centi-seconds, - chans is the channels available, and - satid is the ID of the satellite which took the measurement.
Processing level of product	<i>Level 3B – orthorectified TOA radiance</i>
Measured Quantity Name	<i>TOA radiance</i>
Measured Quantity Units	<i>$W\ sr^{-1}\ m^{-2}\ \mu m^{-1}$</i>
Stated Measurement Quality	<i>From quarterly data quality report:</i> - Coastal Blue: 8.013% - Blue: 6.798% - Green_i: 6.244% - Green_ii: 5.363% - Yellow: 5.771% - Red: 6.277% - Red-edge: 8.736% - NIR: 9.229%
Spatial Resolution	<i>3.7 m – 4.1 m</i>
Spatial Coverage	<i>$\pm 81.5^\circ$ (dependent on season)</i>
Temporal Resolution	<i>Daily at nadir</i>
Temporal Coverage	<i>Mid-March 2020 onwards</i>
Point of Contact	<i>Planet</i>

Product locator (DOI/URL)	<i>Available for purchase through the Planet website</i>
Conditions for access and use	<i>Requires commercial licensing</i>
Limitations on public access	<i>Yes (see above)</i>
Product Abstract	<i>Extracts from Planet content online:</i> <i>The PlanetScope satellite constellation consists of multiple launches of groups of individual satellites. Therefore, on-orbit capacity is constantly improving (in capability or quantity) as technology improvements are deployed at a rapid pace.</i> <i>Each PlanetScope satellite is a CubeSat ~3U form factor (10 cm by 10 cm by 30 cm). The complete PlanetScope constellation of approximately 130 satellites is able to image the entire land surface of the Earth every day (equating to a daily collection capacity of 200 million km²/day). This capacity changes based on the number of satellites in orbit and throughout the season, as satellites image less in the northern hemisphere in the winter time because of a decrease in the amount of hours with sunlight.</i> <i>A third generation of PlanetScope sensors (known as SuperDove or PSB.SD) is currently in orbit and is producing daily imagery with 8 spectral bands (coastal blue, blue, green 1, green, red, yellow, red edge and near-infrared). These satellites were launched in early 2020 and started producing imagery in mid-March 2020. PSB.SD PlanetScope satellites reached near daily cadence in August 2021. Starting from April 29, 2022 all new PlanetScope images have 8-bands and are derived from the PSB.SD sensors (SuperDoves) 8-Band Planet Scope images can be obtained using all Planet Platforms, Integrations and API. The item-type is PSScene.</i>

Availability & Accessibility	
Grade: Basic	
Justification	• <i>Good accessibility and availability, however it does not comply with FAIR.</i> • <i>Description of data management plan is not publicly available.</i>
Compliant with FAIR principles	<i>No.</i>
Data Management Plan	<i>Not known.</i>
Availability Status	<i>Commercially available.</i>

Product Format	
Grade: Good	
Justification	• <i>Metadata and flags provided are good. Not clear if standard naming-conventions used.</i> • <i>Not CEOS-ARD.</i>
Product File Format	<i>Basic Analytic: Data stored as .tiff file, and metadata provided in a .xml file. Informative file naming, although not clear if this is standard.</i>
Metadata Conventions	<i>XML metadata files follow [RD-7] specifications.</i>
Analysis Ready Data?	<i>No</i>

User Documentation		
Grade: Good		
Justification	<i>Several technical documents available for users, including information on the radiometric calibration. Up-to-date information can usually be found via the Planet website, although found in various places. Product ATBD not available to users.</i>	
<i>Document</i>	<i>Reference</i>	<i>QA4ECV Compliant</i>
Information related to PUG	<i>[RD-3, RD-8, RD-5, RD-6]</i>	<i>N/A</i>
ATBD	<i>Not seen</i>	
Radiometric calibration information	<i>[RD-9]</i>	<i>N/A</i>

4 Metrology

Sensor Calibration & Characterisation	
Grade: Good	
Justification	<i>Information into the in-orbit calibration is documented. Quarterly results are published following regular characterisation methods.</i> <i>Pre-launch information including SRF characterisation and calibration is not available to users. High-level overviews have been communicated in public documents, e.g., technical workshops.</i>
References	• [RD-9]

Geometric Calibration & Characterisation	
Grade: Good	
Justification	<i>Information on geometric calibration available, including mention of the use of a DEM and sensor model. Not much information beyond that.</i>
References	• [RD-5]

Metrological Traceability Documentation	
Grade: Not assessable	
Justification	<i>No information found.</i>

Uncertainty Characterisation	
Grade: Basic	
Justification	<i>Proxy for radiometric “uncertainty” obtained through comparisons with five RadCalNet sites. Does not follow an approach based on the GUM [RD-10].</i>
References	• [RD-2] • [RD-10]

Ancillary Data	
Grade: Good	
Justification	<i>DEM used in the processing chain – Information on the DEMs used stated in quarterly report.</i> <i>Atmospheric correction carried out using MODIS input data.</i> <i>SRFs of Planet channels not provided publicly.</i>
References	• <i>[RD-2]</i> • <i>Metadata from files</i>

5 Product Generation

Calibration Algorithm	
Grade: Excellent	
Justification	<i>ATBD-type information defining the calibration algorithm processing steps is provided – appears fit-for-purpose in terms of the mission’s stated performance for all expected use cases.</i>
References	• [RD-9] • [RD-11]

Geometric Processing	
Grade: Good	
Justification	<i>ATBD-type information defining the geometric processing steps is available – appears fit-for-purpose although not explicitly detailed. More information required for assessment.</i>
References	• [RD-11]